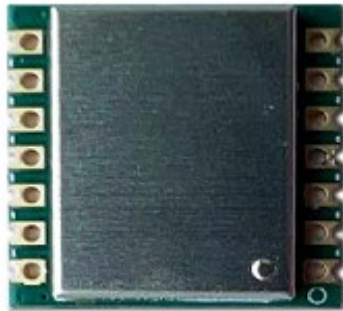
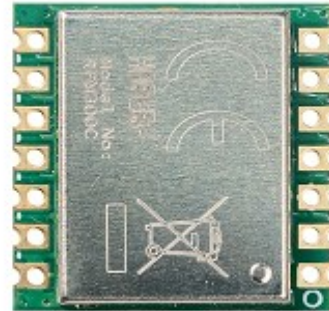


RFM300CW

ISM Transceiver Module With +13dBm Output Power



RFM300CW-915



RFM300CW-433/868

(The purpose of this RFM300CW spec covers mainly for the hardware and RF parameter info of the module, For software info please refer to CMT2300A chip datasheets and demo program of HopeDuino Development Kit)

1. General Introduction

RFM300CW module series' design is based on the high performance CMOSTEK NextGen RF CMT2300A chip, It operates at 433/868/915MHz ISM band, The low receive sensitivity (-118dBm) coupled with $+13\text{dBm}$ output power ensures extended range and improved link performance.

2. Features

- 140dB maximum link budget.
- Low RX current of 7mA.
- $+13\text{ dBm}$ output power.
- Programmable bit rate up to 300 kbps @FSK 40 kbps OOK.
- High sensitivity: down to -118dBm (433.92MHz).
- FSK, GFSK, and OOK modulation.
- SMD Package (16x16x1.8mm).

3. Application

- Meter Reading
- Wireless data collection
- Automobile security system
- Home automation and security system

4. Pin Definition

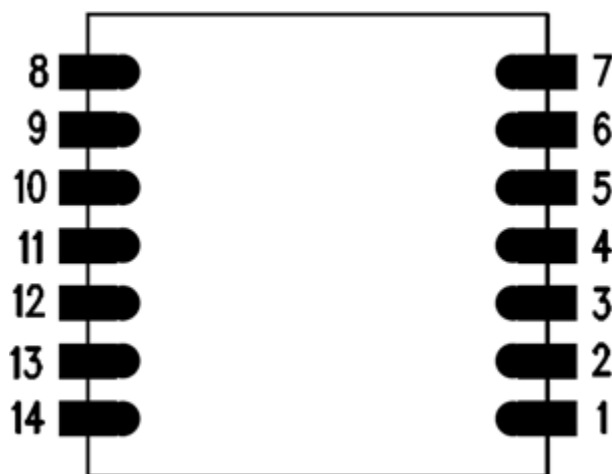


Figure 1. RFM300CW Pin Definition

No.	Definition	Type	Function
1	ANT	AI/ AO	RF signal input/output.
2	3.3V(VDD)	PI	Power supply input, 1.8-3.6V.
3	GND	G	Ground.
4	NC	--	No Connect.
5	CSB	I	SPI Chip select input, active low.
6	SCK	I	SPI Clock input.
7	FCSB	I	SPI FIFO select input, active low.
8	SDIO	I/O	SPI Data input and output.
9	GPIO1	I/O	General Purpose Digital I/O that may be configured through the registers to perform various functions
10	GPIO3		
11	GPIO2		
12	NC	--	No Connect.
13	NC	--	No Connect.
14	GND	G	Ground.

Notes:

[1]. INT1 and INT2 are interrupts. DOUT is demodulated output. DIN is a modulation input. DCLK is a modulation or demodulation data rate synchronization clock, automatic switching in TX/RX mode.

[2]. The SCLK pin connects an internal pull-down resistor (about 4.7 kΩ) inside the chip. Thus, in low-power applications, the MCU cannot output high level (pull up), otherwise it will generate leakage current and will cause low-power implementation failure.

[3]. The SDIO pin connects an internal pull-up resistor (about 15 kΩ) inside the chip. Thus, in low-power applications, the MCU cannot output low level (pull down), otherwise it will generate leakage current and will cause low-power implementation failure.

[4]. The GPIO pins connect an internal pull-up resistor (about 15 kΩ) inside the chip. Thus, in low-power applications, the MCU cannot output low level (pull down), otherwise it will generate leakage current and will cause low-power implementation failure.

5. Electrical Parameter

Parameter	Min.	Max.	Unit
Positive Power Supply	-0.3	+3.6	V
Voltage On Digital Control Inputs	-0.3	VDD + 0.3	V
Voltage On Analog Inputs	-0.3	VDD+ 0.3	V
RX Input Power	-	+10	dBm
Storage Temperature	-55	+125	°C
Soldering Temperature(10s)	-	+255	°C
ESD Rating (Human Body Model)	-2	2	KV

Recommended Working Range

Parameter	Min.	Max.	Unit
Positive Power Supply	+1.8	+3.6	V
Working Temperature	-40	+85	°C
Supply Voltage Slew Rate	1	-	mV/us

DC Characteristic

Parameter	Conditions	Min.	Typ.	Max.	Unit
RFM300CW TX Working Current	433MHz band, Pout=+13dBm	-	29	45	mA
	868MHz band, Pout=+13dBm	-	31	45	
	915MHz band, Pout=+13dBm	-	31	45	
RFM300CW RX Working Current	433MHz band	-	7	10	mA
	868MHz band	-	7.5	10.5	
	915MHz band	-	7.5	10.5	
RFM300CW Sleep Current	All band	-	-	1	uA

Transmitter AC Characteristic

Parameter	Conditions	Min.	Typ.	Max.	Unit
RFM300CW Output Power	433/868/915MHz band	-	+13	-	dBm
Symbol Rate, FSK Mode	Programmable	0.1	-	300	kbps
Symbol Rate, OOK Mode	Programmable	0.1	-	40	kbps
Frequency Deviation, FSK	Programmable	1	-	200	KHz
Frequency Resolution		-	24.8	-	Hz

Receiver AC Characteristic

Parameter	Conditions	Min.	Typ.	Max.	Unit
RX Sensitivity	433MHz	-	-118	-	dBm
FSK Mode FDEV = 10 kHz,	868MHz	-	-116	-	
SR =0.6 kbps,	915MHz	-	-116	-	
Receiver Bandwidth		50		500	KHz
Blocking Immunity	+/-1MHz offset	-	52	-	dB
	+/-2MHz offset	-	74	-	
	+/-10MHz offset	-	75	-	
Image Rejection Ratio	IF=280KHz	-	35	-	dB

6. Typical Application

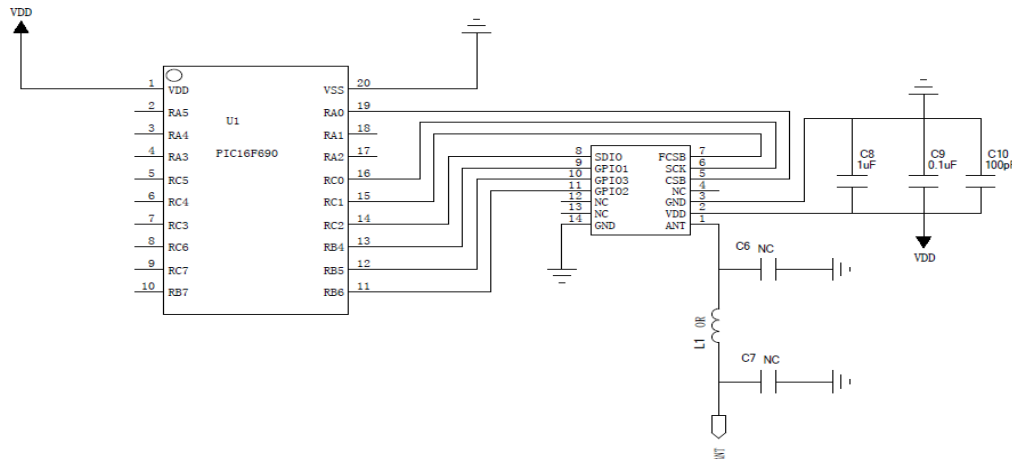
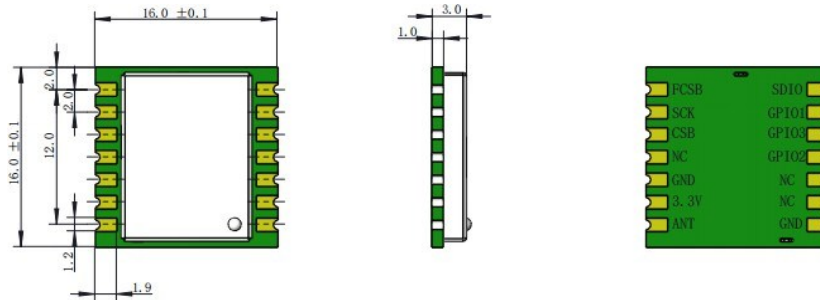


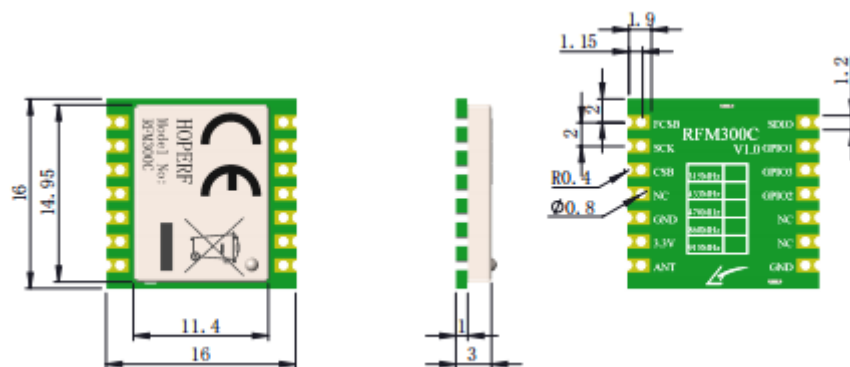
Figure 2. RFM300CW Application

(For software info please refer to CMT2300 chip datasheets and demo program of HopeDuino Development Kit)

7. Mechanical Dimension (Unit: mm)



RFM300CW-915

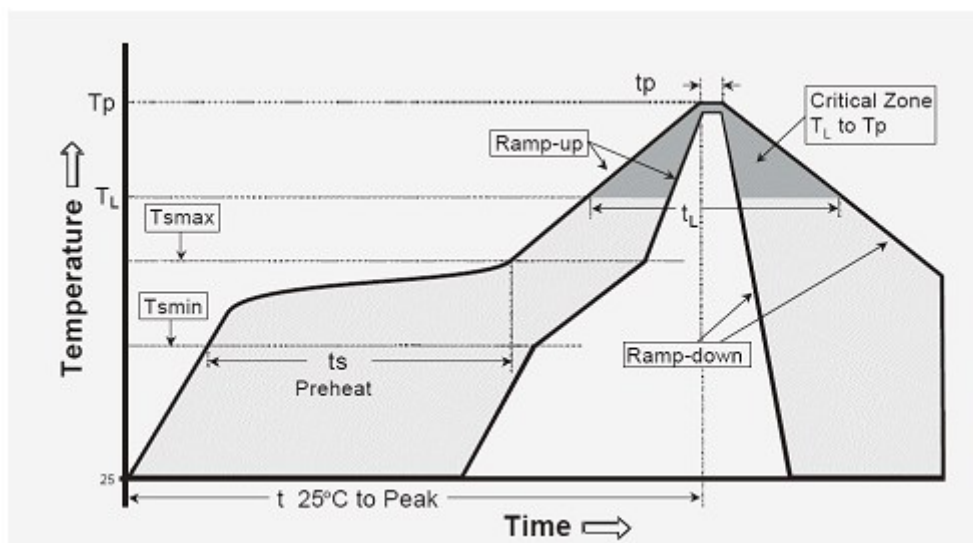


RFM300CW-433/868

Figure3. RFM300CW Mechanical Dimension

8.Reflow soldering temperature curve

It is recommended to comply with the related IPC standards and depend on the actual curve of the solder paste to setup the reflow soldering temperature curve.



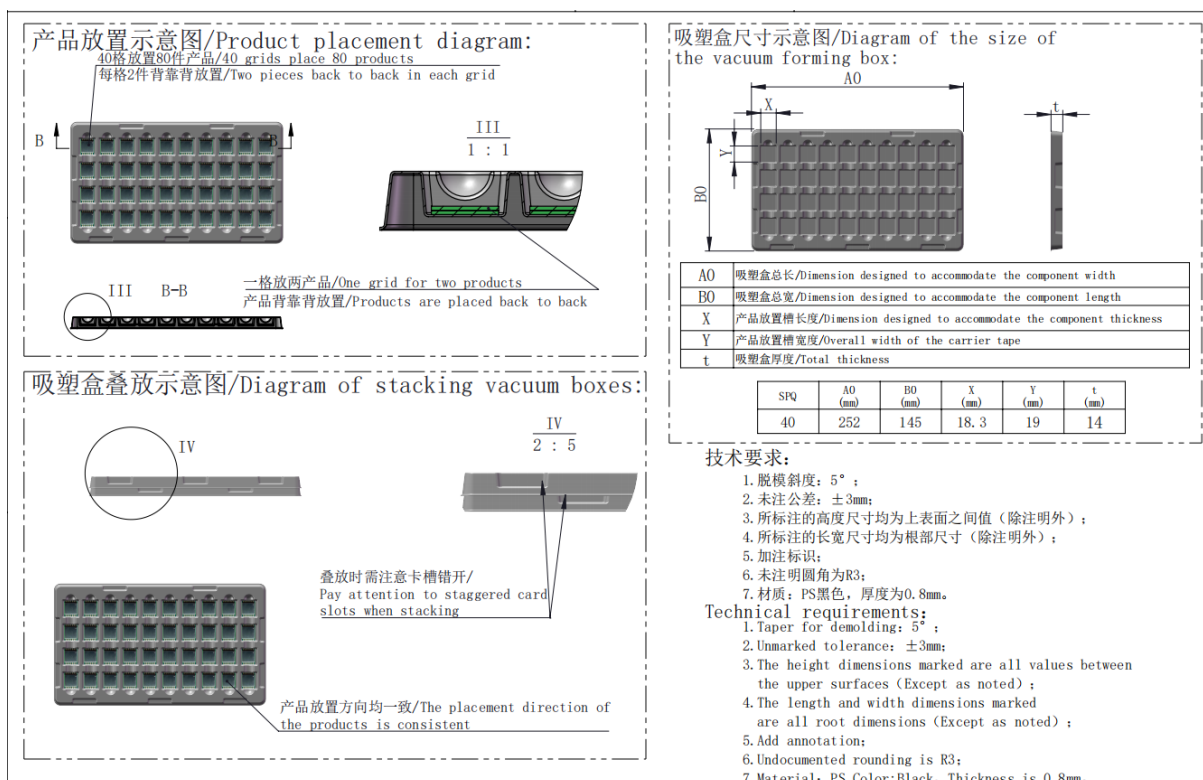
Pre-heating temperature	
Minimum temperature (Tsmmin)	150 °C
Maximum temperature (Tsmmax)	200 °C
Warm-up time (ts)	60 ~ 180 sec
Temperature ramping rate for preheating to the peak temperature (TL-TP)	3 °C/sec (maximum)
Soldering time	60 ~ 90 sec
Soldering temperature	217 °C
Time to reach above the melting point	60 ~ 150 sec
Peak temperature	240 +/- 5 °C
Soldering time for peak temperature +/- °C	10~30 sec
Cooling rate	6 °C/sec (maximum)
Full soldering time	8 min (maximum)

Notes

- The wettability of lead-free solder materials is worse than that of tin-lead eutectic materials, and the melting zone temperature is higher. To achieve better soldering results and protect the solder joints from oxidation, use nitrogen protection during reflow soldering.
- For double-sided products, avoid reflow soldering with the module facing downward (that is, the module should be soldered during the second reflow soldering) to prevent the components on the module from falling off or shifting.
- The module needs to be fully tinned, and it is recommended to use a stepped steel mesh with a thickness of 0.15MM or above.

- When the module must be re-soldered, we recommend manual soldering to avoid possible damage to the module caused by multiple reflow soldering processes.
- Customers should develop welding processes that comply with the above tips based on actual production conditions.

9. Packaging Information



10. Order Information

Model	Frequency	Output Power
RFM300CW-433S2	433MHZ	+13dBm
RFM300CW-868S2	868MHZ	+13dBm
RFM300CW-915S2	915MHZ	+13dBm

11.Contact Information

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